GLOBAL AI HACKATHON 2026 · TEAM STACKOVERFLOW

BestBefore

A self-healing semantic cache for Bifrost. Stale AI answers are rewritten with today's verified facts before they are cached, and every cached answer gets a best-before date.

Statement 04 · Open-Source Bifrost Gateway + Statement 06 · Stale-Answer Detector (Tavily)

github.com/AlgoMasterArpit/HackCulture

Apache 2.0

THE PROBLEM

Models answer from the past. Caches replay it.

USER, 11 NOV 2026

What is the latest stable version of Python?

GEMINI 2.5 FLASH

The latest stable version of Python is 3.12.1.

WIKIPEDIA, FOUND BY TAVILY

"As of October 2026(Python 3.15 is the latest stable release." (verbatim, page markup included)

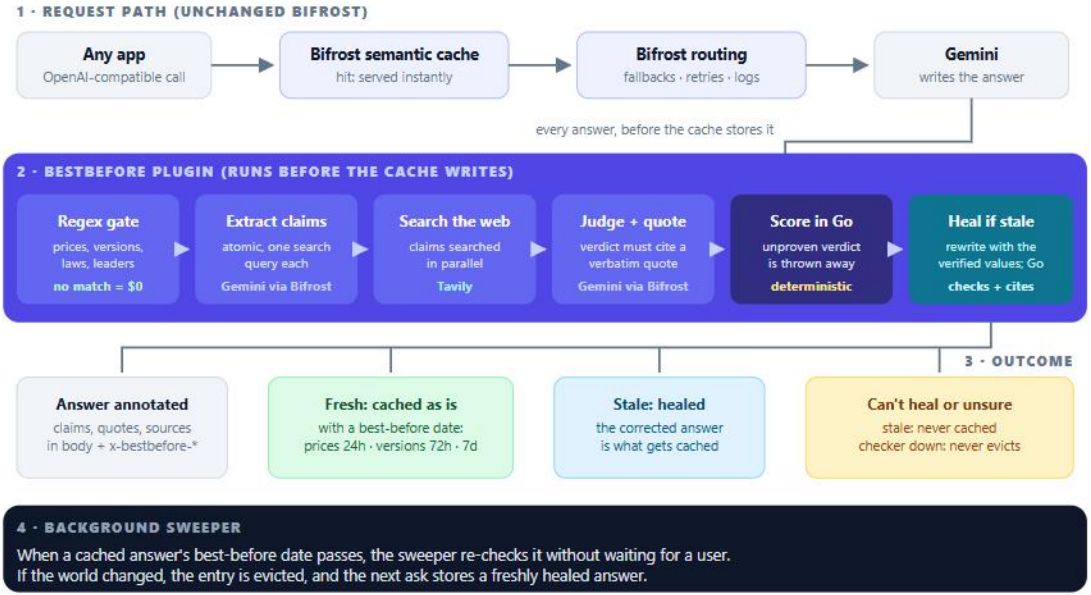
A gateway's semantic cache then serves that answer to **every** user who asks again, until the TTL runs out.

In Bifrost today, the clock is the only thing that removes a cached answer: `isExpiredEntry` checks `expires_at` and nothing else (`plugins/semanticcache/search.go:252`), and expired entries are only cleaned up lazily (issue #2414).

Fact-checkers stop at "this looks outdated". The next user still gets the stale answer.

WHAT WE BUILT

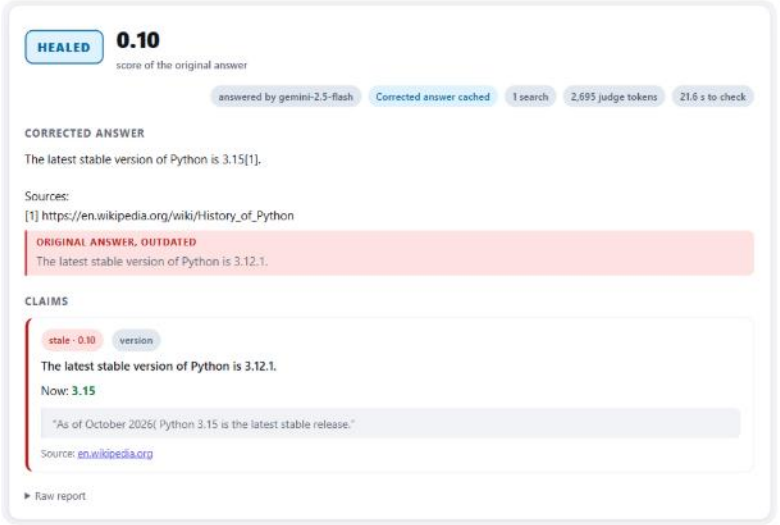
Heal the answer before the cache stores it.



Bifrost runs our post-hook before the cache's, so the cache stores the corrected text

WORKING PROTOTYPE, LIVE

From "3.12.1" to "3.15 [1]", cached for everyone.



Gemini answered from its training data. BestBefore found the claim, Tavily found Wikipedia, the judge quoted it word for word, and Go accepted the rewrite only because it carries the new value, drops the old one and adds no new words or links.

The corrected answer, with its source, is what Bifrost cached. **Asking again: a cache hit with the corrected answer in 1.1 s, no model call.**

RESULTS · LIVE SEMANTIC CACHE

What users actually received.

17 / 20

stale answers delivered by a TTL-only Bifrost cache

0 / 20

with BestBefore; **9** repeat asks got the corrected answer straight from cache



RESULTS · 60 HAND-LABELLED ANSWERS, LIVE GEMINI + TAVILY

Catches stale answers without crying wolf.

80%

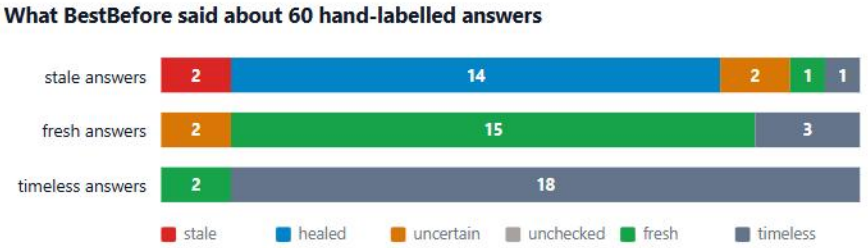
of stale answers caught

0%

of good answers wrongly flagged; **100%** of flags were right

88%

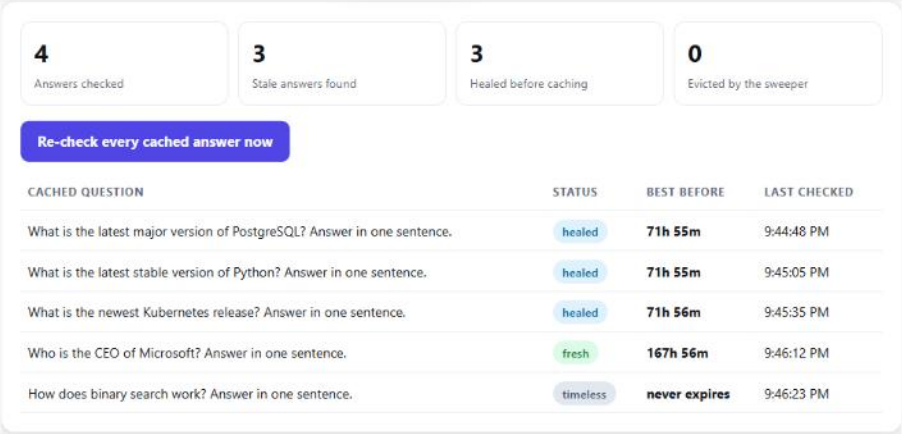
of caught answers healed with the verified value



From eval/run.py. Grounding rule tuned on this set after a first run that caught **65%**; all runs published.

BEST-BEFORE DATES

Every cached answer knows when to be re-checked.



Prices 24h, versions 72h, leaders and laws 7 days. A background sweeper re-checks answers when their date passes, before any user asks, and evicts them if the world changed. Proven end to end: change a price on the mock web, and the sweeper evicts the cached answer on its own.

Cache freshness tab · bestbefore/status

WHY YOU CAN TRUST IT

The model drafts. Go decides.

Verdicts need proof

Every verdict needs a word-for-word quote from the cited source, and a contradiction must name a new value that is in that quote and not in the claim.

Rewrites are checked

A correction may reuse only words from the answer and the verified values, must drop every old number and may not add links. Go appends the sources itself.

Fails safe

Can't prove it or can't heal it: the stale answer is flagged and never cached. Checker down: nothing is evicted.

Broken on purpose, twice

Two adversarial AI reviews found 25 problems, six high-severity, including a phishing link that would have been cached. Each is fixed with a test.

HOW EACH TECHNOLOGY CONTRIBUTES

Take either away and nothing works.

TECHNOLOGY	WHAT IT DOES IN BESTBEFORE
Bifrost plugin order	`PostLLMHook` runs before the semantic cache writes, so the cache stores the healed answer; `semantic_cache-no_store` keeps unhealable stale answers out.
Bifrost transport hooks	See every answer, cache hits included; serve `bestbefore/verify`, `status` and `sweep` inside the OpenAI API.
Bifrost cache API	`cache_id` ties answers to entries; `DELETE /api/cache/clear/{id}` evicts what went stale.
Bifrost routing	Judge and healer calls go back through the gateway: fallbacks across three free Gemini models, retries, logs.
Tavily search	Live evidence per claim: URL, page text, date, and a relevance score that weights the verdict.

HONEST LIMITS · WHAT'S NEXT

What it does not do yet.

- The first answer to a stale question is slower (p95 check **21.2s**); repeats come from cache.
- Streaming responses pass through unchecked.
- The sweeper can evict but not pre-write: Bifrost has no cache-insert API for plugins.
- Best-before dates live in memory, per gateway process.
- Upstream PR drafted in Bifrost's template, building on proposal #9748.
- Store best-before dates in the vector store so replicas share them.
- Check streams after they finish.
- Require two sources on different hosts before healing.

RUN IT

Three commands.

```
git clone https://github.com/AlgoMasterArpit/HackCulture.git
cp .env.example .env     # add Gemini + Tavily keys
docker compose up -d --build
```

Team stackoverflow · github.com/AlgoMasterArpit/HackCulture